

LINUX PROGRAMMING: ASSIGNMENT-4

1. A system has a file /etc/passwd. How would you use grep + tee to extract usernames and save them to a file while also displaying them on screen?

❖ bash

```
grep -o '^[^:]*' /etc/passwd | tee usernames.txt
```

❖ bash

```
cat usernames.txt
```

2. A binary isn't found in \$PATH. How would you use commands (which, find, locate) to troubleshoot and fix the issue?

❖ bash

```
which
```

❖ To find

```
bash
```

```
sudo find / -type f -name <binary_name> 2>/dev/null
```

❖ To locate

```
bash
```

```
locate <binary_name>
```

❖ To fix this trouble shoot we use

```
bash
```

```
export PATH=$PATH:/opt/myprogram/bin
```

3. . Write a command pipeline that finds all .log files modified in the last 24 hours in /var/log and saves results into log_report.txt.

❖ bash

```
find /var/log -type f -name "*.log" -mtime -1
```

```
tee log_report.txt
```

4. What is the difference between shutdown -r now and reboot?

Features	shutdown -r	Reboot
Purpose	Reboot after the shutdown	Reboots the system immediately
Graceful shutdown	Sends warning messages to users, before shutdown.	Doesn't warn the user before reboot.
Complexity	Used when the user want safe controlled reboot	Used for quick reboot
Safety	Safer, ensures file system consistency	Less safety than shutdown -r
Scheduled reboot	Can schedule the reboot for specific time.	Can't schedule

5. . How can you use the tee command to debug a script that generates both standard output and error messages?

❖ bash

```
./myscript.sh | tee output.log
```

❖ bash

```
/myscript.sh 2>&1 | tee debug.log
```

6. Explain any three real-world applications of Linux in industries.

The three real world applications of Linux in industries are:

❖ Web Servers and Hosting:

- Linux is the dominant OS for web servers because it is stable, secure, and open-source.
- Has high traffic efficiently.
- Cost-effective
- Example: Apache, Nginx, and Lighttpd

❖ Cloud Computing and Virtualization:

- Linux powers the majority of **cloud infrastructure** and virtual environments.
- Lightweight and customizable.

- Supports large-scale automation and orchestration.
 - Example: AWS EC2 instances, Google Cloud, and Azure Linux VMs.
- ❖ Embedded Systems and IoT Devices:
- Linux is widely used in embedded systems due to its flexibility and small footprint.
 - Supports custom kernels.
 - Real-time capabilities for industrial automation.
 - Example: Smart TVs, routers, industrial robots.

7. Differentiate application, system and utility software in the context of Linux environment.

Features	Application Software	System Software	Utility Software
Defination	To perform specific tasks for user.	It manages the hardware and software resources of the system	To perform the maintenance and optimization task on the system.
Purpose	Helps users accomplish work	Ensure the system runs correctly	Helps to monitor analyze.
User-interaction	Directly interacts with end user	Mostly interacts indirectly via commands	Can be used directly by users or via commands
Dependency	Runs on top of system software	Works with hardware to support all software.	Depends on system software tp function'
Example	LibreOffice Firefox	Linux kernel Shell	Top/ htop Tar, gzip

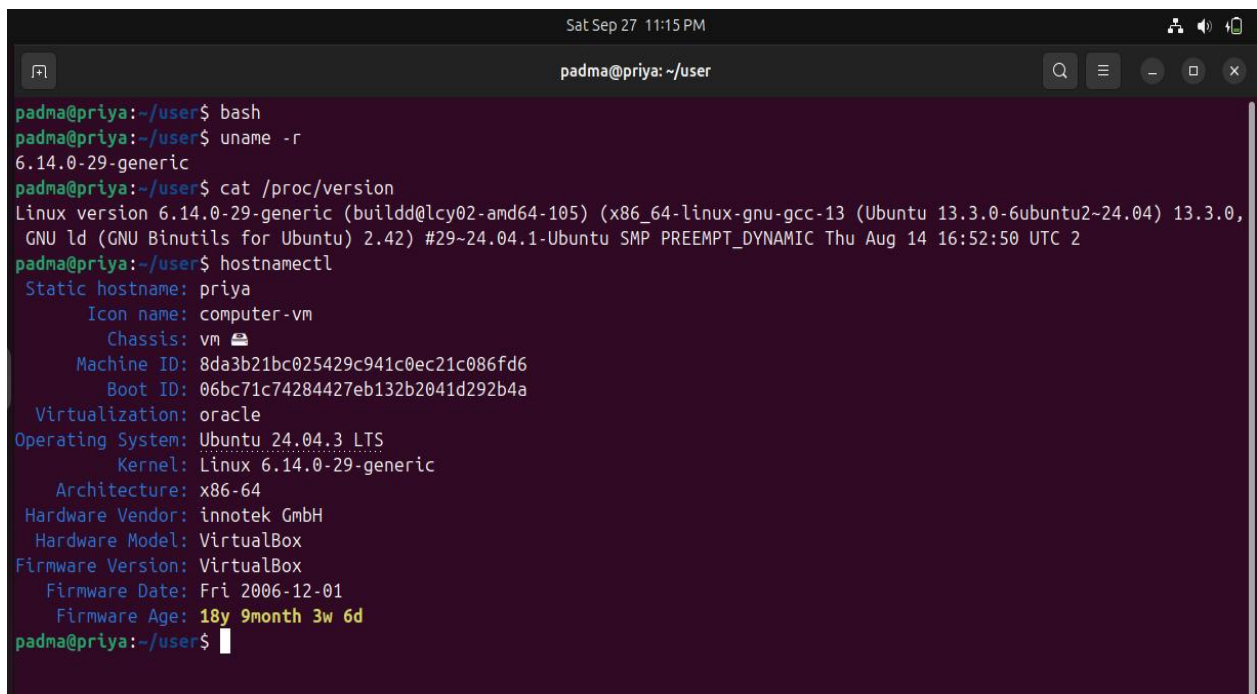
8.What are the key differences between open-source and proprietary operating systems?

Features	Open-source OS	Proprietary OS
Source Code Availability	Source code is free available to view, modify	Source code is closed, users cannot see or modify
Cost	Free or low cost	Paid , license fee is required
Customization	Highly customization	Limited customization
Security	Transparent security	Vendor-dependent
Updates	Frequent and user controlled, you can compile latest versions.	Updates controlled by vendor, schedule may be fixed.

9. Write the command to display the system's kernel version

❖ bash

uname -r



```
Sat Sep 27 11:15 PM
padma@priya: ~/user

padma@priya:~/user$ bash
padma@priya:~/user$ uname -r
6.14.0-29-generic
padma@priya:~/user$ cat /proc/version
Linux version 6.14.0-29-generic (buildd@lcy02-amd64-105) (x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0,
GNU ld (GNU Binutils for Ubuntu) 2.42) #29~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Aug 14 16:52:50 UTC 2
padma@priya:~/user$ hostnamectl
  Static hostname: priya
        Icon name: computer-vm
        Chassis: vm
        Machine ID: 8da3b21bc025429c941c0ec21c086fd6
        Boot ID: 06bc71c74284427eb132b2041d292b4a
        Virtualization: oracle
Operating System: Ubuntu 24.04.3 LTS
        Kernel: Linux 6.14.0-29-generic
        Architecture: x86-64
        Hardware Vendor: innotek GmbH
        Hardware Model: VirtualBox
Firmware Version: VirtualBox
        Firmware Date: Fri 2006-12-01
        Firmware Age: 18y 9month 3w 6d
padma@priya:~/user$
```

10. What is the difference between head and tail commands in text processing?

Features	Head	Tail
Purpose	Displays the beginning of the file	Displays the end of the file.
Syntax	head [option] filename	tail [option] filename
Use case	Quick look at file header Eg: column names in CSV	Quick look at file footer Eg: logs or latest entries
Updates	Usually not used for dynamic files.	-f option allows real-time monitoring of growing files.

❖ bash

head -n 5 file2.txt

❖ bash

tail -n 5 file2.txt

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